Complete user guide

FORGE ONE User Guide

Felucca 1.6.1 and FORGE 0.9 on the M-VAVE FM-1



FORGE ONE · Felucca 1.6.1 + FORGE 0.9 · M-VAVE FM-1
github.com/kkasemsa/FORGE-ONE (forge branch) · 22 October 2026
Felucca by Hügelton Instruments and contributors · GPL-3.0-only

Felucca 1.6.1 with FORGE 0.9 · M-VAVE FM-1

This guide covers the whole instrument: the Felucca firmware by Hügelton Instruments, and FORGE, the motif generator this fork adds. Felucca's own sections follow its official manual, the Felucca wiki, in the order and with the labels the FM-1 itself uses. FORGE has its own chapter (chapter 9) and its own short guide.

Values given as settings are starting points. Judge them by ear.

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1. Installing and going back

FORGE ONE is installed with Felucca's command-line installer, because the official web installer only installs stock Felucca. The steps are in the Read Me First. In short:

1. Download `forge-one.fwsc` from the fork's Releases page (or from the newest green Actions build).
2. Quit every app that uses MIDI. Connect the FM-1 with a USB data cable.
3. Install `mido` and `python-rtmidi` in a Python environment, then run the installer (exact commands in the Read Me First).

4. Wait for the FM-1 to restart. The start-up screen says **FORGE ONE** and its version; MENU > SYSTEM > ABOUT shows FORGE ONE (by Khantipol) above Felucca's credit. Hold SEQ: the SEQ TOOLS map shows FORGE on F5, STARTER on G5.

Stock Felucca is installed from the web installer in Chrome or Edge: connect the FM-1, open the page, press **Install**. No extra hardware is needed. You can also try Felucca in your browser without an FM-1.

Going back to the M-VAVE firmware: use the web installer's **Return to official V15**, or M-VAVE's updater.

Before any install, make a full backup with the web editor. The backup holds projects, presets and settings, but not the FM6 voice bank: keep its `.syx` file.

Installing firmware is at your own risk. If an install is cut off, see chapter 19.

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2. The panel

Area	Controls	What they do
Left knobs	**MASTER**	Volume (speaker, headphones, and USB audio unless MENU > USB LEVEL is FIXED)
	SELECT	Tempo (BPM). MENU > CONTROL > SELECT can make it step pages, play the mod wheel or set key velocity
	PRESETS	The selected track's sound, on every page that is not a list
	ALGORITHM	Selects track T1–T4, on every page
Right knobs	**KNOB 1–4** (K1–K4)	The four values shown on the screen's knob cards
Buttons, row 1	**FX, SCL, ENV, LFO, EDIT, GLO**	Page buttons. Tap for the page, tap again for the next page. FX, SCL, EDIT and GLO also have a quick layer when held
Buttons, row 2	**HOME, SAVE, ARP, SEQ**	Page buttons. HOME held opens the menu, SAVE held is undo, SEQ held is SEQ TOOLS
	PLAY/STOP	Starts and stops all four tracks. Lit green while playing
	REC	Arms the selected track to record. Held: the REC layer
Left of the keys	**OCT−, OCT+**	Octave down / up. On action pages and in dialogs, OCT+ does it and OCT− goes back
Keys	27 keys, **F3 to G5**	16 white keys and 11 black keys. They play the selected track, and in layers they are shortcuts

The screen. The header shows the selected track, the project slot its pattern came from ("1 A", or "1 A*" once edited), the play state, a metronome and the tempo. Below it are the four knob cards, then the page's picture (oscilloscope, piano roll, envelope …), then a footer with hints.

The keys' names. This guide calls the keys by note name: the white keys are F3 G3 A3 B3 C4 D4 E4 F4 G4 A4 B4 C5 D5 E5 F5 G5, and the black keys F#3 G#3 A#3 C#4 D#4 F#4 G#4 A#4 C#5 D#5 F#5. "Black keys 1–4" are F#3, G#3, A#3 and C#4.

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3. Basic gestures

Gesture	What it does
Tap a page button	Opens its page; tap again for the next page of that family. Page buttons act when you let go
Tap HOME	Back to the HOME screen. On HOME, taps go HOME → MIXER → CLOCK
Hold a page button (FX, GLO, SCL, EDIT, REC, SEQ)	Opens its **quick layer**: the keys and knobs become shortcuts and a map shows (chapter 10)
Double-tap a layer button	Keeps the layer open without holding. Tap the button again to close it
Hold **HOME**	Opens the menu (chapter 17)
Hold **SAVE**	Undo the last load or edit. Hold again to redo
Hold **LFO** and turn a knob	**MOMENTARY**: the value jumps back when you let go of LFO. OCT+ while holding keeps it
OCT+ / OCT−	Octave up / down. On action pages and in dialogs: OCT+ does it, OCT− goes back. In the menu: change the value
OCT− + OCT+ together	Octave reset. Held for 5 s: UBOOT, the chip's boot mode for updates (a countdown shows from 2 s; let go to cancel)
GLO + PLAY	Restart from the top without stopping

What can be pressed **breathes** softly, rather than blinking. Idle buttons and keys glow dim so you can find the panel in the dark (MENU > DISPLAY > LEDS).

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4. Tracks and sounds

- **Four tracks**, T1–T4. Each is one synth part with its own engine, sound and 64-step pattern. Drums are a track with the DRUM engine.
- **8 voices** are shared between the four tracks.
- **ALGORITHM** selects the track on every page. The pages and the keys edit and play the selected track.
- **PRESETS** changes the selected track's sound on every page that is not a list. Loading a sound never touches the track's pattern.
- **Thirteen engines** (chapter 11), each with factory presets. Every factory sound has a category (BASS, LEAD, PAD, PLUCK, KEYS, DRUM, FX, OTHER) to browse by.
- **User presets:** 32 slots. **Projects:** 4 slots, A–D (chapter 15).
- **INIT** loads the engine's first preset. **CLEAN INIT** loads its basic sound with nothing added: no FX sends, INSERT off, no LFO, ENV or MOD amounts, no glide, POLY voice mode.
- **RESTORE LAST** (on by default): turn the FM-1 off and on and the music is as you left it.

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5. The pages, button by button

Tap a button for its first page, and again for the next. The knob cards show K1–K4 in order.

Button	Pages (taps go round)	Held
HOME	HOME, MIXER, CLOCK	The menu
SEQ	STEP, DETAIL, AUTOMATION	SEQ TOOLS, on every page
GLO	SONG	The GLO layer
SAVE	USER, PRESETS, PHRASES, PROJECT, TOOLS	Undo
LFO	LFO, LFO 2, LFO DEST, MOD	MOMENTARY (with a knob)
EDIT	EDIT 1, EDIT 2, (engine pages), VOICE, VOICE 2, VOICE 3	The EDIT layer
FX	FX, SLICER, INSERT, INSERT 2, DLY, REVERB, CHORUS	The FX layer
SCL	SCL, CHORD	The SCL layer
ENV	ENV, ENV DEST	
ARP	ARP, ARP 2	
REC	(arms the track)	The REC layer

Knobs on each page

Page	K1	K2	K3	K4
HOME	The selected track's four main sound values (with MENU > HOME LEVELS: T1–T4 LEVEL)			
MIXER	LEVEL	PAN	REV (send)	MUTE
CLOCK	BPM (40–240)	SWING	CLOCK source: INT, USB, TRS	
STEP	The cursor step		A note's LEN (ties)	
DETAIL	CHANCE (and conditions)	RATCH	NUDGE	VEL
AUTOMATION	The row (or PRESETS)	Its step	Its parameter	Its value
SONG	SECTION	TRACK	PAT (slot)	REPS
USER	SLOT	K2–K4 pick LOAD, ERASE or SAVE; OCT+ does it		
PRESETS	Browse (as PRESETS)	ENGINE	FAV	LIST
PHRASES	PHRASE			
PROJECT	SLOT	ERASE	LOAD	SAVE (turn a knob to pick the action; OCT+ does it)

Page	K1	K2	K3	K4
TOOLS	CLEAR SEQ	INIT (one click on: CLEAN INIT)		(OCT+ does it)
LFO	RATE	WAVE: SIN, TRI, SAW, SQR, S&H	PHS (phase)	FADE
LFO 2	SYNC: OFF, 4BAR … 1/32	TRIG: NOTE, FREE	POL: BI, UNI	
LFO DEST	PIT	FLT	SHP	AMP
MOD	SLOT (1–4)	SRC	DST	AMT
EDIT 1 / EDIT 2	The engine's E1–E4 / E5–E8			
FILTER (ANALOG)	TYPE: LP, BP, HP	CUT	RES	KTR
LANES / LANES 2 (DRUM)	KICK, SNARE, CLAP, HATCL / HATOP, TOM, RIM, BELL levels			
VOICE	VCE: POLY, MONO, LEG, UNI	GLIDE	GLMOD: RATE, TIME	PRIO: LAST, LOW, HIGH
VOICE 2	ALLOC: ROT, REUSE	DTUNE	PAN	MUTE
VOICE 3	SPRD (stereo spread)			
FX	DST (distortion)	CHO send	DLY send	REV send
SLICER	OFF, GATE, STUT	PAT (1–16)	RATE	DEPTH
INSERT	TYPE	A	B	C (named by TYPE)
INSERT 2	MIX (dry / wet)			
DLY (all tracks)	TIME	FDBK	COLR	MIX
REVERB (all tracks)	TYPE: ROOM, SPRING, HALL	SIZE	DAMP	
CHORUS (all tracks)	RATE	DEPTH		
SCL	ROOT	SCL	QNT	TRN
CHORD	CHRD	VOIC		
ENV	ATK	DEC	SUS	REL
ENV DEST	FLT	PIT	SHP	ESYNC
ARP	MODE	RATE	OCT	GATE
ARP 2	SWG	PROB	HOLD (latch)	ORD: NOTE, PLAY

- **EDIT's engine pages** come after EDIT 2: FILTER on ANALOG; LANES and LANES 2 on DRUM; SLICES on SLICE; OPERATOR, OP ENV and OPERATOR 2 on FM6.
- **DLY, REVERB and CHORUS** are shared by all tracks. Each track sends to them from its FX page.

6. The sequencer

Each track has a 64-step pattern with its own **LEN** (1–64), **DIV** (4BAR … 1/32, triplets included), **SWING** and **GATE**. Set these four in the SEQ TOOLS layer (hold SEQ, K1–K4).

A step holds up to 4 notes (a chord), or is a TIE (it holds the note before) or a REST. It can also have: **accent**, **slide**, **CHANCE** or a **condition**, **RATCH** (played 2, 3 or 4 times in its length), **NUDGE** (timing) and **VEL**.

6.1 Entering notes

- **Step recording.** On SEQ > STEP with the transport stopped, press **REC**, then play notes: each key (or chord) goes into the cursor step and the cursor moves on. MENU > CONTROL > CHORD ENTRY picks how chords are entered: HOLD (keys held together become the step) or ADD (each key adds a note, K1 moves on).
- **Live recording.** Press REC on any other page. If the transport is stopped, it starts playing too. Play, and the notes are recorded as the loop goes round (overdub). Each note keeps your timing as a NUDGE.
- **The piano roll.** SEQ > STEP draws the pattern. K1 moves the cursor, K3 sets a note's length (right adds ties, left removes them). When the pattern is longer than one screen, the roll follows the play position while playing.
- **The drum grid.** On a DRUM track, SEQ > STEP is a grid: white keys are steps, black keys 1–8 pick the lane (KICK, SNARE, CLAP, HATCL, HATOP, TOM, RIM, BELL). Steps 1, 5, 9 and 13 are marked.
- **STEP PREVIEW** (MENU > CONTROL): moving the cursor while stopped plays the step it lands on.

6.2 DETAIL: chance, ratchet, nudge, velocity, conditions

SEQ > DETAIL is the STEP page with K1–K4 as the cursor step's **CHANCE**, **RATCH**, **NUDGE** and **VEL**, drawn as lanes over the steps. PRESETS walks the steps; on a DRUM track a white key picks the step. **EDIT** puts the step's four values back to their defaults.

Conditions. Turn K1 past 100 % to choose a condition instead of a chance: **1:2 2:2 1:3 2:3 3:3 1:4 2:4 3:4 4:4 FILL !FILL**.

- **a:b** plays the step on loop *a* of every *b* loops of its track: 1:2 every other loop, 4:4 only every 4th. Each track counts its own loops, from PLAY and from the start of each SONG section.
- **FILL** steps play only while **GLO + E4** is held; **!FILL** steps only while it is not.
- A step has a chance or a condition, not both.

6.3 Timing: NUDGE and QUANTIZE

Live recording keeps when you played each note as a per-step **NUDGE**, in 1/16 of a step (from half a step early to 7/16 late). The track's **QUANTIZE** decides what plays: ON (the default) plays every step on the grid, OFF plays your own timing. Set nudges by hand on DETAIL (K3), or hold steps on STEP and turn PRESETS.

6.4 Parameter locks and automation

- **Locks.** On SEQ > STEP, hold a step and turn K1–K4: those values sound on that step only. The knobs lock the parameters of the sound page you showed last (HOME, ENV, LFO, FX, EDIT …). **EDIT** with the step held clears its locks.

- **Automation.** Knob moves can be recorded per step while recording. An automation icon marks the knob cards it drives.
- **SEQ > AUTOMATION** lists everything that happens on the track's steps: PLAY / CLEAR, QUANTIZE, each step's CHANCE, RATCH, NUDGE and VEL, its locks and its automation. Change a row's step, parameter or value, turn a lock into automation or back, add or delete rows (EDIT deletes). There are 128 records shared by the four tracks.

6.5 Metronome and count-in

CLICK (OFF, while recording, or always) at three levels, and a **COUNT-IN** of 1 or 2 bars before recording from stop. Set them in the REC layer or MENU > AUDIO. The click goes to the headphones and speaker only, never to USB audio or into a pattern.

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7. SEQ TOOLS

Hold **SEQ** on any page. The white keys become pattern tools for the selected track, and K1–K4 are **LEN**, **DIV**, **SWING** and **GATE**. Each tool is one undo (SAVE held). **OCT−** puts the track back as it was when the layer opened.

Key	Tool	What it does
F3	CLEAR	Clears steps 1 to LEN
G3	REVERSE	Plays the pattern backwards
A3 / B3	SHIFT < / >	Moves every step one left / right
C4	RANDOM	A new beat (DRUM), or new notes in the scale on the same rhythm
D4	COOK	Changes the pattern a little; press again for more
E4	BEAT	DRUM only: four on the floor, backbeat, 8th hats
F4	EUCLID	A Euclidean rhythm: K1 HITS, K2 STEPS, K3 ROT. OCT+ keeps it, OCT− cancels
G4 – C5	Lane CLEAR, REVERSE, FILL, RANDOM	DRUM only: the selected lane (black keys 1–8 pick it)
D5	DOUBLE	The pattern twice (LEN ×2, up to 64 steps)
E5	HALF	The first half (LEN ÷ 2)
F5	**FORGE**	FORGE ONE: a seeded motif generator, a melody on a synth track or a beat on a DRUM track (chapter 9). While it is open, C4 and D4 reroll its two halves
G5	**STARTER**	FORGE ONE 0.6: all four tracks at once, a sound and a pattern each (bass, lead, pad, drums), from one seed (9.6). OCT− puts all four back

Tools only work while no SONG is playing ("STOP TO EDIT").

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8. Songs

GLO > SONG chains up to 16 **sections**. Each section names the project slot (A–D) that **each track** plays, or "-" for a track that rests there, and how many times it repeats. The page draws a timeline with a lane per track.

- K1 SECTION, K2 TRACK, K3 PAT, K4 REPS.
- The white keys A3, B3, C4 and D4 set the slot.
- Each section starts every track's loop count again, so conditions work per section: in a section repeated 4 times, a 4:4 step plays on its last repeat.

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9. FORGE

FORGE writes a musical idea onto a track in one key press, then lets that idea evolve: a melody on a synth track, a beat on a DRUM track (9.5). Every pattern comes from a **seed**: the same seed and settings always give the same steps. The FORGE User Guide has more recipes and detail.

9.1 Using it

1. Select a track. On a synth track set its ROOT and SCALE (SCL page) and LEN (SEQ TOOLS K1) first.
2. Hold **SEQ**, tap **F5**. The pattern is written at once (heard while playing) and drawn as a grid of steps: bright cells are notes with a pitch mark, half-bright cells are ties, dark cells are rests.
3. Turn the knobs. Each turn rewrites the pattern from the steps as they were before FORGE opened.
4. **OCT+** (or F5 again, another tool, or letting go of SEQ) keeps it. **OCT−** cancels.
5. **SAVE** held afterwards undoes the whole FORGE, if no other tool ran since.

Knob	Parameter	Range	What it does
K1	SEED	0–99	Picks the pattern
K2	DENSITY	10–100 %	How many of the style's steps sound (above 75 %, extra notes too)
K3	EVOLVE	0–4	How much the idea changes across the pattern and from loop to loop
K4	STYLE	8 styles	Rhythm and pitch profile. Any turn sets DENSITY to the style's default
C4	New notes	NOTES 0–99	The same rhythm, new notes
D4	New rhythm	RHYTHM 0–99	A new rhythm; each step that still sounds keeps its pitch (at EVOLVE 0)

The header briefly shows `FORGE <seed> <style>` after each turn, and the screen shows the C4 / D4 counts under the steps. K1 SEED sets both counts back to 0. F5 is lit while FORGE is open.

9.2 Styles

Style	Motif	Rhythm	Pitch	Octave
BERLIN	8	16th ostinato	Small steps around the root	C3

Style	Motif	Rhythm	Pitch	Octave
DETROIT	16	Syncopated stabs	4-note scale chords	C4
DUB	16	Off-beats, long ties	3-note scale chords	C3
MELODIC (card: MELO)	16	8ths	Rises then falls in an arc	C4
ACID	16	16ths, accents, slides	Root pedal, octave jumps	C2
TRANCE	8	Rolling 16ths	Triad arpeggios	C4
SYNTH	8	8ths	Root, fifth, octave	C3
AMBIENT (card: AMBI)	16	Very sparse, long ties	Wide intervals, some chords	C4

Step 1 is always the root. Every note is in the track's ROOT and SCALE; chords are built from real intervals kept in the scale, so they work in every scale (minor in CHR). Melodies move above and below the root; octave jumps only go up.

9.3 EVOLVE

EVOLVE	What happens
0	A pure loop: every copy of the motif is the same; no chance, conditions or ratchets
1	Later copies sometimes sit a scale step away; a few ornament notes get a 75 % or 50 % chance; ACID and TRANCE ratchet a few
2	More drift; some ornaments get a condition (2:2, 3:4, 2:4, 4:4)
3	As 2, and later copies drop or add a note
4	The most change

The first copy keeps the motif's notes and rhythm (it can still get chances, conditions and ratchets). With LEN equal to the motif length there is only one copy.

9.4 What it writes

Steps 1 to LEN of the selected track: notes, NOTE / TIE / REST, accent, slide, ratchet, chance or condition, and velocity (louder on the beat, softer on off-beat 16ths). It does not change steps past LEN, the track's LEN, DIV, SWING or GATE, its sound, or the other tracks. Applying removes locks from steps that end up empty. Projects made with FORGE open on stock Felucca 1.6 and 1.6.1.

9.5 On a DRUM track

The same knobs and style names make a beat: K1 SEED, K2 DENSITY (the style's main hits stay from 50 % up, extra hats over 75 %), K3 EVOLVE (0 a pure loop; 1 and up moves hats and adds ghost snares with a chance; 2 and up ends the last bar in a snare-roll fill on a 4:4 condition; 3 and up moves kicks off the beat), K4 STYLE. **C4** rerolls the low lanes (kick, snare, clap, tom) and **D4** the hats, rim and bell. The screen shows each step as eight strips, KICK at the bottom, an accent brighter (at LEN 33 to 64, two columns: kick, snare, clap, tom on the left; hats, rim, bell on the right). The kick on step 1 is always there. The result is ordinary drum-grid steps.

Style	Beat
BERLIN	Four on the floor, straight hats, rim ticks, no snare
DETROIT	Four on the floor, clap on 2 and 4, open hats off the beat
DUB	Kick on 1 and 9, rim on 2 and 4, sparse hats
MELODIC	Four on the floor, clap on 2 and 4, off-beat hats
ACID	Four on the floor, snare on 2 and 4, off-beat 16th hats, ratchets
TRANCE	Four on the floor, clap on 2 and 4, open hats off the beat, dense
SYNTH	Kick on 1 and 7, heavy snare on 2 and 4, 8th hats
AMBIENT	A kick on 1, sparse hats, tom, rim and bell

9.6 STARTER: all four tracks (G5)

Hold **SEQ** and tap **G5** on any track. STARTER loads a sound on each track and writes a pattern for each from one seed: **T1 BASS, T2 LEAD, T3 PAD, T4 DRUM** (T4 becomes a DRUM track). All four take the selected track's ROOT; the SCALE is yours if you set one, or the starter's if the track is on CHR. The tempo goes to the starter's BPM, unless the FM-1 follows an external clock.

K1 SEED, K2 DENSITY (50–150 % of each part's own), K3 EVOLVE and K4 GENRE (13 genres). Black keys 1–4 pick the genre's variation (3 or 4 each, 45 starters in all); each loads its own sounds, tempo, scale and swing. Sounds in *italics* are from the **FORGE PACK** (0.9): 40 sounds built on factory presets with their own settings (the full list is in the FORGE User Guide); they save and load as ordinary edited sounds, and SAVE > USER keeps one. C4 rerolls the notes of all four, D4 the rhythms. **OCT+** (or G5 again) keeps it; **OCT−** puts all four tracks back as they were, sounds, steps, automation and tempo included. SAVE-held undo cannot undo STARTER, so use OCT− before you let go of SEQ. LEVEL, PAN and MUTE are never changed.

Genre (K4)	Black key, variation	T1 bass	T2 lead	T3 pad	T4 kit · beat	BPM	Scale*	Swing
Berlin	1 **SEQ**	*BERLIN SEQ*	TRIO ARP LEAD	*BERLIN PAD*	DRUM DEEP KIT · Berlin	120	MIN	–
	2 **DRIFT**	LOFI WAVE BASS	*BERLIN RESO*	*STRING MACH*	DRUM PERC KIT · Ambient	108	MIN	–
	3 **CHOIR**	FM6 FM BASS	TRIO SYNC LEAD	*CHOIR PAD*	DRUM DEEP KIT · Berlin	116	HARM	–
	4 **ARP**	*DEEP BASS*	LOFI STEP LEAD	*BERLIN PAD*	*RHYTHM BOX* · Berlin	124	DOR	–
Detroit	1 **STABS**	FM6 FM BASS	*WARM STAB*	FM6 TINE EP	*HOUSE KIT* · Detroit	126	DOR	–
	2 **STRNG**	*DEEP BASS*	ANALOG STRINGS	*HOUSE PIANO*	*HOUSE KIT* · Detroit	128	MIN	–
	3 **ELECT**	*ELECTRO BASS*	LOFI PULSE LD	PHASE STRING	*BOOM KIT* · Electro	125	PHRY	–
	4 **DEEP**	FM6 FM BASS	*GLASS ARP*	ANALOG SOFT PAD	DRUM DEEP KIT · Melodic	124	MIN	–

Genre (K4)	Black key, variation	T1 bass	T2 lead	T3 pad	T4 kit · beat	BPM	Scale*	Swing
Dub Techno	1 **CHORD**	*DUB SUB*	*DUB CHORD*	GRAIN SHIMMER	*DUB KIT* · Dub	118	DOR	–
	2 **DEEP**	VOICE WOW BASS	FM6 PAD	GRAIN CLOUD PAD	*DUB KIT* · Dub	120	MIN	–
	3 **MINI**	*DUB SUB*	PHASE RESO	*DUB PAD*	*TECHNO KIT* · Dub	124	DOR	–
	4 **ECHO**	ANALOG SUB BASS	FM6 TINE EP	GRAIN FROZEN	*DUB KIT* · Dub	116	MIN	–
Melodic	1 **ARC**	TRIO FAT BASS	ANALOG SAW LEAD	FM6 PAD	DRUM DRUM KIT · Melodic	122	MIN	–
	2 **ARP**	*DEEP BASS*	TRIO SYNC LEAD	*TRANCE PAD*	DRUM TIGHT KIT · Melodic	124	MIN	–
	3 **PIANO**	FM6 FM BASS	SAMPLE PIANO	GRAIN CLOUD PAD	DRUM DEEP KIT · Melodic	122	DOR	–
	4 **PLUCK**	TRIO FAT BASS	*TRANCE PLUCK*	*CHOIR PAD*	DRUM DRUM KIT · Melodic	123	MIN	–
Acid House	1 **CLASS**	*ACID 303*	*HOUSE PIANO*	ANALOG SOFT PAD	*HOUSE KIT* · Deep house	124	MIN	10%
	2 **JACK**	*ACID SQUARE*	ANALOG RAVE	VOICE CHOIR AAH	*HOUSE KIT* · Deep house	122	DOR	15%
	3 **FLOOR**	*ACID 303*	LOFI ARP 8BIT	PHASE ORGAN	DRUM DRUM KIT · Acid	128	MIN	–
Acid Techno	1 **ROLL**	ANALOG ACID	*CHIP ARP*	ANALOG SOFT PAD	*TECHNO KIT* · Acid	128	MIN	–
	2 **WARE**	*ACID SQUARE*	LOFI STEP LEAD	ANALOG PWM STR	*TECHNO KIT* · Acid	135	MIN	–
	3 **HARD**	*ACID 303*	FM6 BRASS	PHASE WIRE	*TECHNO KIT* · Acid	140	PHRY	–
Trance	1 **GATE**	ANALOG SUB BASS	TRIO ARP LEAD	*TRANCE PAD*	*HOUSE KIT* · Trance	138	MIN	–
	2 **LIFT**	ANALOG SUB BASS	*SUPERSAW*	*CHOIR PAD*	*HOUSE KIT* · Trance	138	MIN	–
	3 **PSY**	ANALOG SQR BASS	TRIO RING BELL	PHASE RESO	DRUM TIGHT KIT · Trance	142	PHRY	–
Deep House	1 **KEYS**	*HOUSE BASS*	FM6 TINE EP	ANALOG SOFT PAD	DRUM DEEP KIT · Deep house	122	DOR	15%
	2 **ORGAN**	VOICE WOW BASS	*HOUSE ORGAN*	ANALOG STRINGS	*HOUSE KIT* · Deep house	120	MIN	20%

Genre (K4)	Black key, variation	T1 bass	T2 lead	T3 pad	T4 kit · beat	BPM	Scale*	Swing
	3 **VOCAL**	ANALOG SUB BASS	*VOX CHANT*	FM6 PAD	DRUM DEEP KIT · Deep house	121	MIN	10%
Synthwave	1 **DRIVE**	*OCTAVE BASS*	ANALOG SAW LEAD	ANALOG PWM STR	*BOOM KIT* · Synthwave	110	MIN	–
	2 **NIGHT**	TRIO FAT BASS	*BRASS STAB*	FM6 PAD	*BOOM KIT* · Synthwave	98	MIN	–
	3 **8BIT**	LOFI WAVE BASS	*8BIT LEAD*	TRIO CHIP CHOIR	DRUM TIGHT KIT · Synthwave	118	MAJ	–
Breakbeat	1 **AMEN**	*REESE*	FM6 BRASS	ANALOG STRINGS	*BREAK KIT* · Breakbeat	134	MIN	10%
	2 **NUSK**	*ACID SQUARE*	LOFI STEP LEAD	PHASE STRING	*BREAK KIT* · Breakbeat	135	PHRY	5%
	3 **ELECT**	*ELECTRO BASS*	*SCREAM*	FM6 PAD	*BOOM KIT* · Electro	128	MIN	5%
Ambient	1 **CLOUD**	*DRONE BASS*	FM6 BELL	GRAIN CLOUD PAD	DRUM PERC KIT · Ambient	80	LYD	–
	2 **DRONE**	PHYS DRONE STRING	SAMPLE FLUTE	*AIR PAD*	DRUM PERC KIT · Ambient	72	DOR	–
	3 **GLASS**	*DRONE BASS*	PHYS MARIMBA	*CHOIR PAD*	DRUM PERC KIT · Ambient	84	LYD	–
	4 **RAIN**	FM6 FM BASS	TRIO RING BELL	*GLASS PAD*	DRUM PERC KIT · Ambient	76	MIN	–
Lo-Fi	1 **PIANO**	*LOFI BASS*	SAMPLE PIANO	*LOFI PAD*	*BOOM KIT* · Lo-fi	84	DOR	25%
	2 **TAPE**	TRIO FAT BASS	*WARM KEYS*	WHEEL SOFT FLUTE	DRUM DRUM KIT · Lo-fi	88	MIN	20%
	3 **SAX**	*LOFI BASS*	SAMPLE SAX	FM6 PAD	DRUM PERC KIT · Lo-fi	80	DOR	22%
Jarre	1 **OXY**	ANALOG SQR BASS	*OXY LEAD*	*STRING MACH*	*RHYTHM BOX* · Rhythm box	104	MIN	–
	2 **EQUI**	LOFI WAVE BASS	TRIO ARP LEAD	VOICE CHOIR AAH	*RHYTHM BOX* · Electro	120	MIN	–
	3 **FIELD**	FM6 FM BASS	*BRASS STAB*	ANALOG PWM STR	*RHYTHM BOX* · Rhythm box	112	DOR	–

Genre (K4)	Black key, variation	T1 bass	T2 lead	T3 pad	T4 kit · beat	BPM	Scale*	Swing
	4 **ZOO**	VOICE WOW BASS	*VOX CHANT*	GRAIN SHIMMER	DRUM TIGHT KIT · Electro	118	PHRY	–

* The starter's SCALE is used only when the selected track's SCALE is CHR (the default). Any other SCALE you set is kept.

10. Quick layers

Hold a button for its layer. Keys pressed in a layer are shortcuts: they play no notes, send no MIDI and record nothing. Double-tap the button to keep the layer open. In SET layers (GLO, SCL, EDIT, SEQ, REC), **OCT−** puts back what the layer changed since it opened.

FX layer (hold FX)

The white keys hold performance effects on the master while pressed. **Black keys 1–4 mute T1–T4** while held. K1–K4 are the macros **FILTER**, **CRUSH**, **THROW** (the dry mix into the delay and reverb) and **DEPTH**.

Key	Effect	Key	Effect
F3	REPEAT 1/8	A4	OCT DN (harmonizer an octave down)
G3	REPEAT 1/16	B4	FLANGER (in time with the tempo)
A3	REPEAT 1/32	C5	PHASER (in time with the tempo)
B3	REVERSE	D5	REV THROW (every track's reverb send full)
C4	LPF	E5	ECHO THROW (every track's delay send full)
D4	HPF	F5	CUTOFF THROW (the master low-pass closes; FORGE ONE 0.7)
		G5	DIST THROW (every track's DIST at least 100; FORGE ONE 0.7)
E4	TAPE STOP		
F4	FREEZE		
G4	OCT UP (harmonizer an octave up; DEPTH is its shimmer)		

- **Any white key can hold any effect:** hold the key in the layer and turn PRESETS. EDIT puts its default back.
- **MENU > CONTROL > FX LATCH** makes the effects toggle on and off, so nothing has to stay held. FX + OCT− turns everything off.
- **The throws flow back (FORGE ONE 0.7).** REV, ECHO, CUTOFF and DIST THROW come in at once, and when you let go they glide back to the tracks' own settings over about 0.75 s instead of stopping.

- **DUCK (FORGE ONE 0.7)** is a sidechain pump: while it is on, every kick of a DRUM track dips the other tracks by 12 dB, and they come back over an 8th note. It is not on a key by default: assign it to a white key with PRESETS, or use PERFORM's D#5.

PERFORM (double-tap FX) — FORGE ONE 0.7

Double-tap **FX** and the FX layer stays open as **PERFORM**: both hands free, mutes, solos, scenes and effects on one screen. The white keys are the FX map's effects, exactly as above. The black keys and knobs change:

Key	What it does
F#3 G#3 A#3 C#4 (black 1–4)	**MUTE** T1–T4, latched (press again to unmute)
OCT+ held + black 1–4	**SOLO** T1–T4, latched; several can be soloed
D#4 F#4 G#4 A#4 (black 5–8)	**Scenes 1–4**: tap to recall the mutes and solos stored there; hold 0.6 s to store the current ones
C#5 (black 9)	**QUANT**: BAR (mutes, solos and scenes wait for the next bar while playing) or NOW
D#5 (black 10)	**DUCK** on / off
F#5 (black 11)	**CLEAR**: every track on, no solo, every effect and macro off, at once
K1–K4	T1–T4 **LEVEL**. Tap **OCT+** alone to switch them to the FX macros (FILTER CRUSH THROW DEPTH) and back
OCT−	Every effect and macro off (as FX LATCH's ALL OFF)
FX (tap)	Ends PERFORM

The screen shows each track as ON, MUTE or SOLO (a track that will change at the next bar is lit in the waiting colour), the scenes (stored ones bright, the last one recalled selected), QUANT, DUCK, the knobs' mode and the effects that are on. Hold a white key to see the effect map, and turn PRESETS to reassign it as in the FX layer.

- **QUANT BAR** puts the change in exactly at the bar line, before the bar's first step plays. Stopped, changes are immediate.
- Mutes are the tracks' MUTE (saved with the project, as GLO's). Solos and DUCK stay after you leave PERFORM, until CLEAR or GLO's C4 UNMUTE ALL. Scenes and QUANT are kept with the settings, not the project.

GLO layer (hold GLO)

Key / knob	What it does
Black keys 1–4	MUTE T1–T4 (latched; lit = sounding)
F3, G3, A3, B3	SOLO T1–T4 while held
C4	UNMUTE ALL
D4	MIDI LEARN on / off (chapter 16)
E4	FILL while held (FILL / !FILL steps)

Key / knob	What it does
F4	TAP tempo
K1–K4	T1–T4 LEVEL
GLO + PLAY	Restart from the top without stopping
GLO + SELECT	Tempo, even with BPM LOCK

SCL layer (hold SCL)

Press any key: its note name becomes the track's **ROOT**. K1–K4 are ROOT, SCL, CHRD and VOIC. The LEDs show the root lit and the scale's notes breathing.

EDIT layer (hold EDIT)

The white keys from F3 pick the **engines**, one key each, in PRESETS order. After them come **INIT** (E5) and **CLEAN INIT** (F5), which ask first. K1 ENG, K2 No. (the engine's sounds), K3 FAV. On a DRUM track the black keys 1–8 **mute its lanes** (KICK … BELL), for live play only: never saved or recorded.

SEQ layer (hold SEQ)

SEQ TOOLS: chapter 7, FORGE on F5 and STARTER on G5: chapter 9.

REC layer (hold REC)

Key	What it does
F3	CLEAR: the selected track's steps and automation (one undo)
G3	CLICK: OFF, REC, ON
A3	COUNT-IN: OFF, 1 BAR, 2 BARS
B3	CLICK LEVEL: LOW, MID, HIGH

K1–K3 set the same three settings. A tap of REC still arms and disarms the track.

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11. Engines

In the order the FM-1 lists them:

Engine	What it is
ANALOG	Virtual analog: two oscillators (WAVE, SYNC, SUB), noise, drive, a resonant LP / BP / HP filter (EDIT > FILTER)
FM6	Classic 6-operator FM (Dexed-based), 32 algorithms. SLOT picks a factory patch (F1–F8), the track's own (OWN) or a voice from a bank sent by the web editor (B1–B32). Edit on OPERATOR, OP ENV, OPERATOR 2, or in the web editor (imports `.syx`)
PHASE	Phase distortion, with a wavefolder (FOLD, EDIT 2)

Engine	What it is
LOFI	Chiptune: pulse, triangle, saw, noise and a 4-bit wave RAM, stepped envelope, sweep, arpeggio
SAMPLE	Multisampled PIANO, FLUTE, SAX, and 3 user sample slots
VOICE	Formant oscillator: sung vowels
TRIO	3 oscillators with ring modulation and sync, multimode filter
WHEEL	Tonewheel organ: drawbars, percussion, key click, drive, rotary speaker
GRAIN	Granular textures from the built-in samples or a user slot
PHYS	Physical models: modal resonators, strings, struck membranes, sympathetic strings
NOISE	Noise from analog to digital: colours, crackle, shift-register and metallic tones
SLICE	A drum break, a piano note or your own sample cut into slices, one per key; set slices by hand on SLICES
DRUM	An 8-lane kit on the General MIDI key map, all synthesized. KIT: STD, 80, 10, 66, 55, 77, DEEP, TIGHT, PERC. EDIT > LANES sets each lane's level

A missing sample (an empty user slot) plays a plain sine at the note's pitch, and the screen says NO SAMPLE once. User samples are uploaded from the web editor.

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12. Effects

Per track, in signal order:

1. **DIST** (FX page, K1): distortion.
2. **INSERT** (FX > INSERT): TYPE OFF, SOFT, HARD, FOLD, FUZZ (drives: DRIVE, TONE, LEVEL), CRUSH (BITS, RATE, LPF), PHASR, FLANG, CHOR (RATE, DEPTH, FDBK). **INSERT 2**: MIX (dry / wet).
3. **SLICER** (FX > SLICER): a tempo-synced 16-step GATE or STUTter with 16 patterns.
4. **Sends** (FX page, K2–K4): CHO, DLY and REV, to the shared effects.

Shared by all tracks:

- **DLY:** TIME (1/4, 1/8, 1/16, 1/32, triplets, 1/2, 1/1, 2BAR, 4BAR, and dotted 1/4D, 1/8D, 1/16D), FDBK, COLR, MIX. In time with the tempo; it holds up to 1.49 s.
- **REVERB:** TYPE **HALL** (default, smooth stereo, up to about 4.5 s), ROOM or SPRING; SIZE; DAMP.
- **CHORUS:** RATE, DEPTH.
- A **master limiter**.

The **FX layer** (chapter 10) adds performance effects on the master.

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13. Modulation

- **ENV** (ENV page): ATK, DEC, SUS, REL. **ENV DEST**: the envelope's amount to FLT, PIT and SHP, and **ESYNC**, which sets ATK, DEC and REL as note values of the tempo (1/64T to 4 bars).
- **LFO** (LFO page): RATE, WAVE (SIN, TRI, SAW, SQR, S&H), PHS, FADE. **LFO 2**: SYNC to a note value (4BAR to 1/32, following the tempo or an external clock), TRIG (NOTE or FREE), POL (bipolar or unipolar). **LFO DEST**: the LFO's amount to PIT, FLT, SHP and AMP.
- **MOD matrix** (LFO > MOD): 4 slots per track. K1 picks the slot, then SRC, DST, AMT.
  - Sources: LFO, ENV, VEL, KEY, RAND, MODW, AT (aftertouch), EXPR, S&H, SLEW.
  - Destinations: PITCH, CUT, SHP, AMP, PAN, DIST, CHO, DLY, REV, RATE, VIB, E1–E8, DEPTH (the LFO's own depth), VEL and TIME (humanize: each note's velocity, a step up to 30 ms late).
- **MOMENTARY:** hold LFO and turn a knob on HOME or a sound / FX page; let go and the value jumps back.
- **SPREAD** (EDIT > VOICE 3): the voices of a POLY or UNISON track alternate left and right of its PAN.

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14. Scales, chord keys and the arpeggiator

SCL page: ROOT, SCL (the scale), QNT, TRN (transpose ±24).

- **Scales:** CHR, MAJ, MIN, DOR, MIX, PEN, MPEN, HARM, PHRY, LYD, LOC, MEL, BLUES, WHOLE, DIMHW, DIMWH.
- **QNT** (how the keys and sequence follow the scale):
  - OFF: the normal chromatic keyboard.
  - SNAP: every key, rounded down to the scale.
  - WHITE: the scale on the white keys only; C4 plays ROOT, black keys are silent.
  - SEQ: as SNAP, and the sequenced notes are snapped as they play (the steps are not changed).
  - SEQ+T: as SEQ, and the sequence follows TRN too (record at TRN 0, then turn it).
- **MENU > SCALE LEDS** shows the scale on the keys all the time.

CHORD page (tap SCL again): **CHRD** makes one finger play a chord: OFF, DIA3 / DIA7 (the scale's triads or sevenths), or fixed shapes MAJ, MIN, DOM7, MAJ7, MIN7, SUS4, POW. **VOIC**: CLOSE, OPEN, INV1, INV2, +OCT. Chord keys work on the keys, MIDI in, recording and the arpeggiator.

ARP page: MODE, RATE, OCT (1–4), GATE. **ARP 2**: SWG, PROB, HOLD (latch), ORD. Modes: UP, DN, UPDN, RND, ORD (the order you pressed), REPEAT, and for chords DNUP, UP+8, CONV, DIVG, PINKY, THUMB, WALK, CHORD. The ARP button flashes on the beat.

Voice modes (EDIT > VOICE): POLY, MONO, LEG (legato), UNI (unison), with GLIDE.

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15. Saving: sounds, patterns and projects

What	Where	How
A sound (user preset)	SAVE > USER	Stop, tap SAVE, pick a slot with K1, then OCT+ and OCT+ again. Name it with the keys. 32 slots, each with a category
Load a factory or user sound	SAVE > PRESETS, or turn PRESETS	K2 ENGINE, K3 FAV (favourites), K4 LIST
A pattern from a project	SAVE > PHRASES	K1 picks the phrase, OCT+ loads it into the selected track
A project (all four tracks, sounds, patterns, song)	SAVE > PROJECT	K1 picks the slot (A–D). Turn K4 to pick SAVE (K3 LOAD, K2 ERASE), then OCT+. EDIT names it. Save while stopped
Clear a sequence, INIT a sound	SAVE > TOOLS	K1 CLEAR SEQ, K2 INIT or CLEAN INIT, OCT+ does it
Undo	Hold SAVE	Undoes the last load or edit; hold again to redo
Everything	Web editor	Full backup and restore (not the FM6 voice bank), and one project to a file and back

RESTORE LAST keeps an autosave of its own, written when the transport is stopped and nothing has been touched for 10 s. The four project slots are never overwritten by it.

16. MIDI and USB

USB is class-compliant: MIDI in and out, and a stereo audio input called "Felucca" (44.1 or 48 kHz) that records the master output on the computer. No driver is needed. On macOS 13–15, set MENU > SYSTEM > USB SERIAL to OFF so the audio input appears.

MIDI in comes over USB or TRS. **MENU > MIDI > MIDI IN** decides which channels play which track:

MIDI IN	Channels
CH1-4 (default)	Channels 1–4 play T1–T4
CH5-8, CH9-12, CH13-16	Another block of four (for example two FM-1s on one cable)
SEL	Every channel plays the selected track
TRACK	Each track its own channel: MENU > MIDI > T1–T4 CHANNEL (1–16, OMNI or OFF)

- The keys send on the selected track's channel. Pitch bend, sustain and panic are supported.
- **Clock** from INT, USB or TRS (HOME > CLOCK, K3).
- **AUTO SELECT:** a MIDI note into another track makes it the selected one.
- **CC map:** 5 GLIDE, 7 LEVEL, 10 PAN, 71 resonance, 72 release, 73 attack, 74 brightness, 75 decay, 91 reverb send, 93 chorus send, 94 delay send.
- **MIDI LEARN** (GLO + D4): turn an FM-1 knob, then move a control on your controller. Up to 16 learned CCs, for any knob of any track. OCT− clears the picked one; MENU > MIDI > LEARN CLEAR clears them all.

- **Program Change** on a track's channel loads that track's sound by its number in PRESETS (from 0). The engine never changes.

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17. The menu

Hold **HOME**. ALGORITHM moves between the five tabs, PRESETS through the rows, and any knob (or OCT+ / OCT−) changes the value. Press HOME to close. The web editor's Settings tab can change all of these too.

Tab	Rows
DISPLAY	COLOR (10 palettes, and CUSTOM from the web editor), STYLE (FLAT, LINE), LARGE, ANIM (ON, OFF, IDLE), LEDS (OFF, DIM LO, DIM HI, INV, INV LO, INV HI), SCREEN OFF (NEVER … 60 MIN), SCOPE (OUT, MIX), HOME (SCOPE, TRACKS, LEVELS)
CONTROL	HOLD (layer hold time), KNOB ACCEL, FX LATCH, BPM LOCK, SCALE LEDS, STEP PREVIEW, CHORD ENTRY (HOLD, ADD), SELECT (BPM, PAGE, MODW, VEL)
AUDIO	SPEAKER EQ (FLAT, LOWCUT, BASS+), USB LEVEL (MASTER, FIXED), CLICK, CLICK LEVEL, COUNT-IN, TUNE (±50 cents)
MIDI	MIDI IN, T1–T4 CHANNEL, AUTO SELECT, SCREEN WAKE (+MIDI, PANEL), LEARN CLEAR
SYSTEM	USB SERIAL, RESTORE LAST, HELP, 2ND CORE (beta), CALIBRATION, INFO, ABOUT

- **SPEAKER EQ** also shapes the headphones and USB audio: keep it on FLAT when recording.
- **HELP ON** shows a short hint when a page or layer opens.
- **2ND CORE** (new in 1.6.1, beta, OFF by default): ON renders some parts on the second core from the next power-on, with the same sound. A unit whose second core misreads goes back to one core and sets it OFF.
- **INFO** shows the version, the USB link and the CPU load.

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18. Playing with other gear

These are setup suggestions for sequencing the FM-1 from a controller, a groovebox or a DAW.

- **Four parts from one sequencer:** set MIDI IN to **CH1-4** and send each part on channels 1–4, or use **TRACK** and give each track the channel your sequencer already uses.
- **Tempo sync:** set HOME > CLOCK source to USB or TRS to follow an external clock. LFO SYNC, ENV SYNC, the delay and the SLICER follow it too. COUNT-IN does not work with an external clock.
- **FORGE with external sequencing:** FORGE writes the FM-1's own patterns. Run the FM-1's sequencer in sync (external clock) for FORGE tracks, and play other tracks from your sequencer over MIDI.
- **Recording:** the USB audio input records the master; set USB LEVEL FIXED for a level independent of MASTER, and SPEAKER EQ to FLAT.

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19. Troubleshooting

Problem	What to do
An install stopped part-way	Run the same install command again (Read Me First): it finishes the write
The screen stays black after an install	Check for a USB device named **WL80UBOOT** (boot mode); the MIDI installer cannot reach it. Try another USB data cable, quit every MIDI app and run the web installer again. If it stays in boot mode, FM-1 Transporter with a Seeed XIAO RP2040 can write the official firmware
The installer cannot find the FM-1	Use a data cable, not a charge-only one. Quit Ableton Live and other MIDI apps
The Mac does not see the USB audio input	MENU > SYSTEM > USB SERIAL OFF (macOS 13–15)
A tool says STOP TO EDIT	A SONG is playing. Stop it first
F5 shows no FORGE	The firmware is stock Felucca, or older than FORGE 0.5 on a DRUM track
G5 shows no STARTER	The firmware is older than FORGE 0.6
A step sometimes does not play	It has a CHANCE or a condition: check SEQ > DETAIL
Chords play one note	Voice mode is MONO, LEG or UNI: set POLY on EDIT > VOICE
A project does not open on an older firmware	See Appendix B

Bugs in Felucca itself: Felucca Issues. Bugs in FORGE or this fork: FORGE ONE Issues.

20. Appendix A: Basic operations

I want to ...	Do this
Pick a track	Turn ALGORITHM
Change a track's sound	Turn PRESETS
Change the tempo	Turn SELECT (or GLO + SELECT, HOME > CLOCK, GLO + F4 tap)
Start / stop	PLAY
Restart from the top while playing	GLO + PLAY
Record live	REC, then play
Enter steps	SEQ > STEP, stop, REC, play notes
Set pattern length	Hold SEQ, turn K1 (LEN)

I want to ...	Do this
Make a random variation	Hold SEQ, D4 (COOK)
Generate a motif	Hold SEQ, F5 (FORGE), turn K1–K4, OCT+
Lock a value on one step	SEQ > STEP, hold the step, turn a knob
Mute a track	Hold GLO, press its black key (1–4)
Play a fill	Hold GLO + E4 (FILL steps)
Throw the reverb	Hold FX + D5 (REV THROW)
Undo	Hold SAVE
Save a sound	Stop, SAVE, K1 slot, OCT+, OCT+
Save a project	SAVE (tap on to PROJECT), K1 slot, turn K4 (SAVE), OCT+
Open the menu	Hold HOME

21. Appendix B: Compatibility between versions

From	To	What to know
FORGE ONE	Felucca 1.6 / 1.6.1	Everything opens and plays the same; FORGE patterns are ordinary steps
1.6 / 1.6.1 (and FORGE)	1.5 / 1.5.1	Projects with **conditional steps** do not open there (FORGE at EVOLVE 2 and up can write them). Clear conditions first, or keep a backup. MIDI IN TRACK plays as a channel block, FOLD is ignored, DEEP / TIGHT / PERC play as 77, MOD VEL / TIME read as DEPTH
1.5 or later	1.2 – 1.4.1	Projects saved by 1.5 or later do not open
1.4 or later	1.1.5.x	Projects saved by 1.4 or later do not open

Every release opens everything older releases saved. Full notes: Felucca release notes.

22. Credits and licence

Felucca is by Leo Kuroshita (@kurogedelic), Hügelton Instruments, with many contributors: see CREDITS.md. Its engines and DSP include work ported from CrispyZebra, DaisySP, Mutable Instruments' eurorack code, Dexed, klattsch and others, each under its own licence (LICENSES/). This guide draws on the Felucca wiki, the official manual.

FORGE and this fork, FORGE ONE, are by the FORGE ONE contributors.

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